Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 32: Correcting for Overlap

The Principle of Inclusion-Exclusion

Chartrand & Zhang, Section 8.2, pp. 280–286

Name: _____ Date: _____

Where we are

The Addition Principle requires the options not to overlap. When they do, adding the two counts counts the overlap *twice*. Today: how to fix that — with a formula you could have derived from a Venn diagram in Week 5.

By the end of today you should be able to

1. Apply inclusion–exclusion for two and three sets.
2. Count the complement when the direct count is harder.
3. Explain the signs in the formula using Venn regions.

1. Two sets

A puzzle. "Name a state capital or one of the 50 largest U.S. cities." How many correct answers are there? Not 100 — Phoenix is both, and so are several others.

Inclusion–Exclusion, two sets

For finite sets A and B:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

If A and B are disjoint, this reduces to the Addition Principle.

Why. Say a elements are only in A, b only in B, and x in both. Then $|A|+|B| = (a+x)+(b+x)$ counts the overlap twice; subtracting $|A \cap B| = x$ counts it once.

Example 8.16. A class has 17 computer science majors, 11 mathematics majors, and 5 in both. Students in CS or mathematics: $17 + 11 - 5 = 23$.

Example 8.15. How many 8-bit strings begin with 110 or end with 1100?

- Begin with 110: $2^5 = 32$
- End with 1100: $2^4 = 16$
- Both — the form 110?1100 with one free bit: 2

Total: $32 + 16 - 2 = 46$.

2. Three sets

Inclusion–Exclusion, three sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Why the signs alternate. Track an element lying in all three sets: it is added 3 times, subtracted 3 times, then added back 1 time — counted once in total. Every Venn region comes out counted exactly once.

Example. How many integers from 1 to 100 are divisible by 2, 3, or 5?

$\lvert A_2\rvert = 50$	$\lvert A_3\rvert = 33$	$\lvert A_5\rvert = 20$
$\lvert A_2 \cap A_3\rvert = 16$	$\lvert A_2 \cap A_5\rvert = 10$	$\lvert A_3 \cap A_5\rvert = 6$
$\lvert A_2 \cap A_3 \cap A_5\rvert = 3$		

Total: $50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$.

Where the intersections came from

An integer divisible by both 2 and 3 is divisible by 6, so $|A_2 \cap A_3|$ counts the multiples of 6 up to 100, which is $\lfloor 100/6 \rfloor = 16$. This works because 2 and 3 share no common factor. (For 4 and 6, the intersection is multiples of 12, not 24.)

3. Counting the complement

How many integers from 1 to 100 are divisible by **none** of 2, 3, 5?

Counting them directly is awkward. Counting the ones we do *not* want is easy — we just did it. So:

$$100 - 74 = 26.$$

Complement counting

If the elements you want are hard to count, count those you do not want and subtract from the total. This is $|\overline{A}| = |U| - |A|$ — Lecture 12 again.

The most common error

Forgetting the $+|A \cap B \cap C|$ term. Subtracting all three pairwise intersections removes the triple overlap entirely, so it must be added back. If your three-set answer is too small, check that term first.

Exit ticket

1. How many integers from 1 to 100 are divisible by 3 or 7?

2. Explain in one sentence why the triple intersection is *added* back.

Before Friday

Read Section 8.3, "The Pigeonhole Principle," pp. 286–292.

Self-check (not collected). Section 8.2, Exercises 1, 3, 5, and 7 — odd-numbered, answers in the back.

Homework 11 is due Friday, November 6, 11:59 p.m.

Questions I still have